Requirements elicitation

Functional requirements

Customer registration: The system should support account creation by users, the users may provide their identification, contact and employment details for registration.

- Register customers and verify identity
- The system should register individual customers with their names,national id, address and phone number
- Register company customers with company name, registration number, contact person address and phone number

User authentication: A system that's capable of ensuring users or registrations are legitimate by checking their credentials and employing multi-factor authentication.

- Login with credentials and multifactor authentication (biometrics, safety keys)
- Access according to roles (customer, admin, manager, teller,auditor)
- Rate limiting of sessions, timeout sessions and lockout after multiple login attempts
- The system should let a user log out and return to the login screen
- A teller must be able to find a user using id number, registration number or name
- Each login is to be linked to one customer record and store each user's role

Account management: overlooking all activities that can happen within an account's capacity. Deposits, transactions, withdrawals and loans are maintained under customer profiles.

- Open, modify, freeze and close accounts
- Maintain user profiles
- The system should provide accounts a system generated account number and record the opening and closing balance
- A customer must be able to hold multiple accounts, of any type
- Investments accounts should require a BWP500 deposit before opening

Transaction history: all transactions that take place using a chosen account are recorded and aligned with their respective date, time as well as references.

- Book all transactions
- Include sender and recipient references
- User should be able to view the transaction history of an account they may access

Transaction: the system should allow ease of money flow in and out of accounts to take place.

- Deposits, withdrawals and balance enquiries
- Transfers between accounts, including own, and other banks
- Bill payments and standing orders
- Transaction reversal
- Transaction limits
- Allow withdrawals only from investment and cheque accounts
- The system must reject transactions larger than the available balance

Interest: Savings earns 0.05% a month and investment 5%, while cheque earns none

- The system shouldn't pay interest twice in the same month
- Interest must be paid to every savings and investment account at once
- Interest should be rounded to the nearest two decimal places, add it to the balance and record it as an interest transaction

Nonfunctional requirements

Security: all passwords should be stored as salted hashes. The system should deny any attempt by a customer to change another customer's details. The system should timeout the user after 3 failed login attempts

Performance: a monthly interest run over 1000 accounts should finish within 30 seconds. Login, deposit, withdraw and account display must respond within 3 seconds

Usability: A new user shall be able to login, open an account and make a deposit fast without help. Every error message should state what the problem is and how it can be fixed

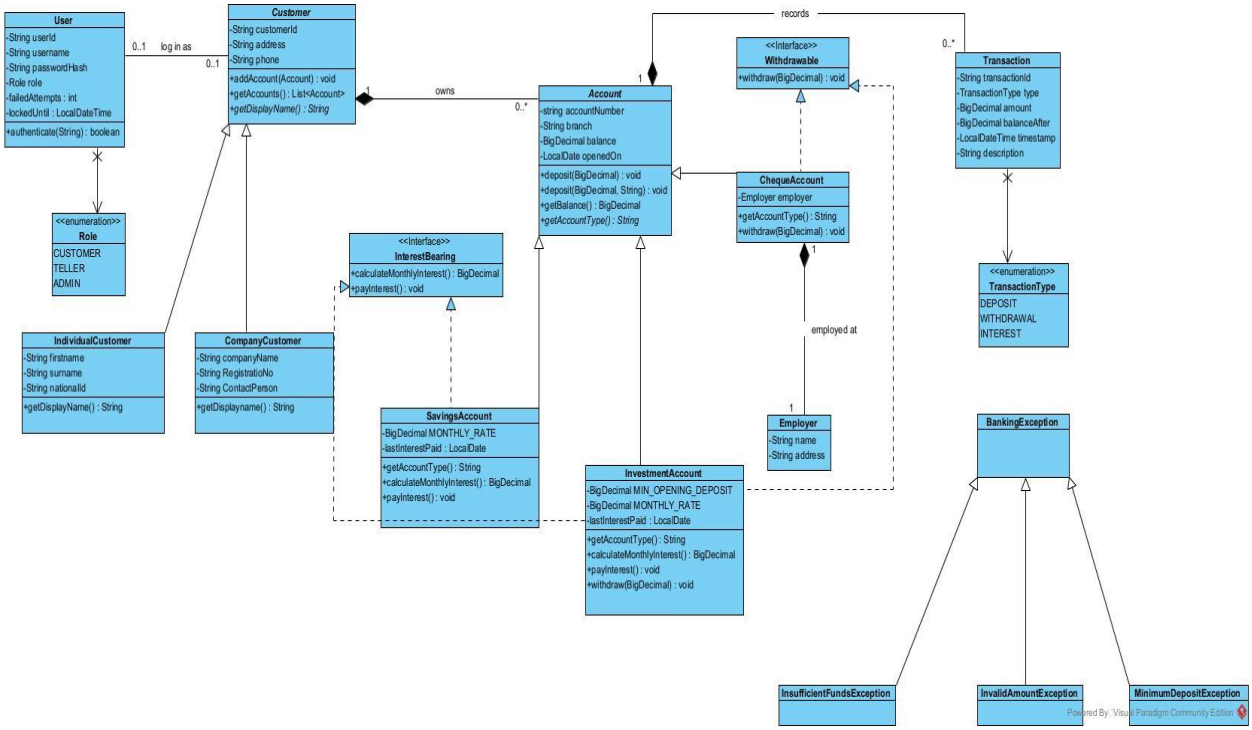
Maintainability: code follows MVC layers, having no SQL in the model or view and no JavaFX imports in the model or controllers

Reliability: invalid input shouldn't crash the system.

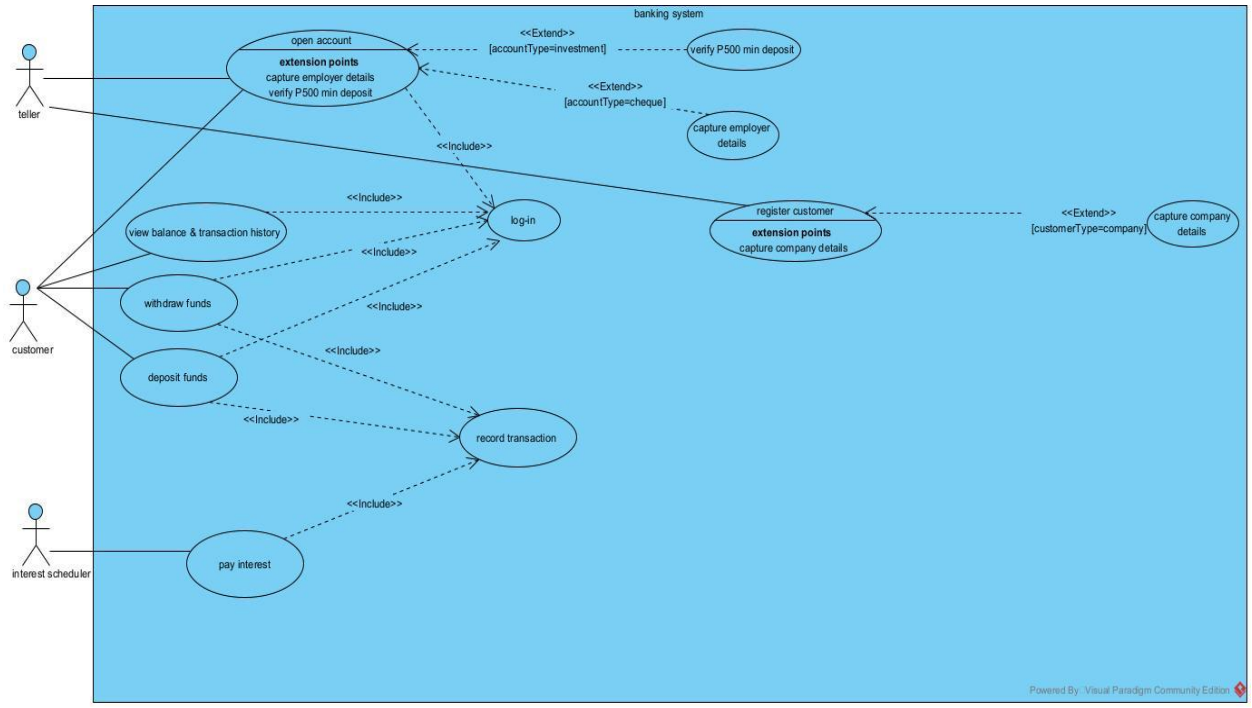
Data integrity: A balance change and its transaction record shall be saved in one database transaction, both happen or neither do. Transaction records shall be append-only. Money is to be calculated as BigDecimal, with no rounding required

UML Modelling

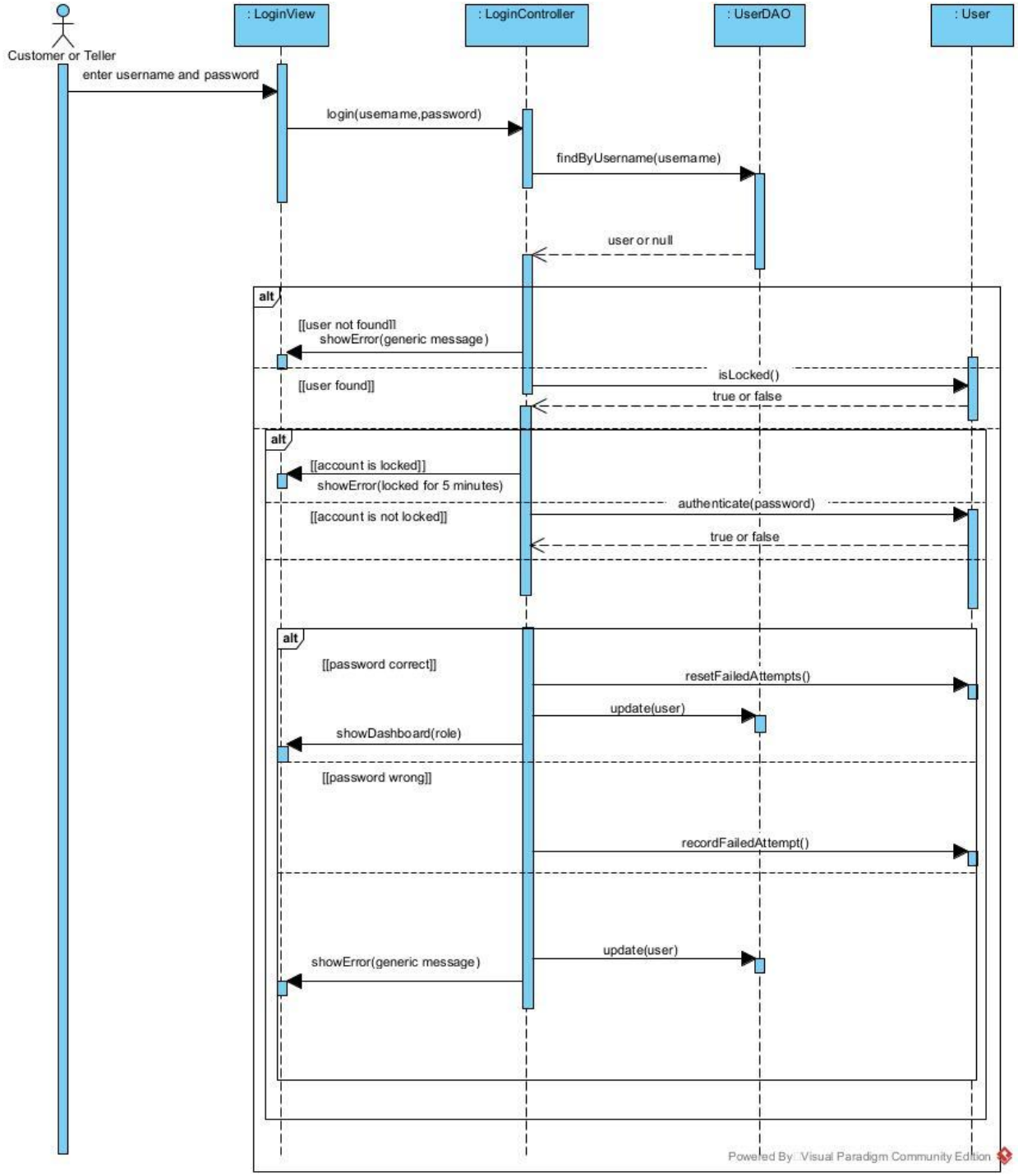
Class diagram



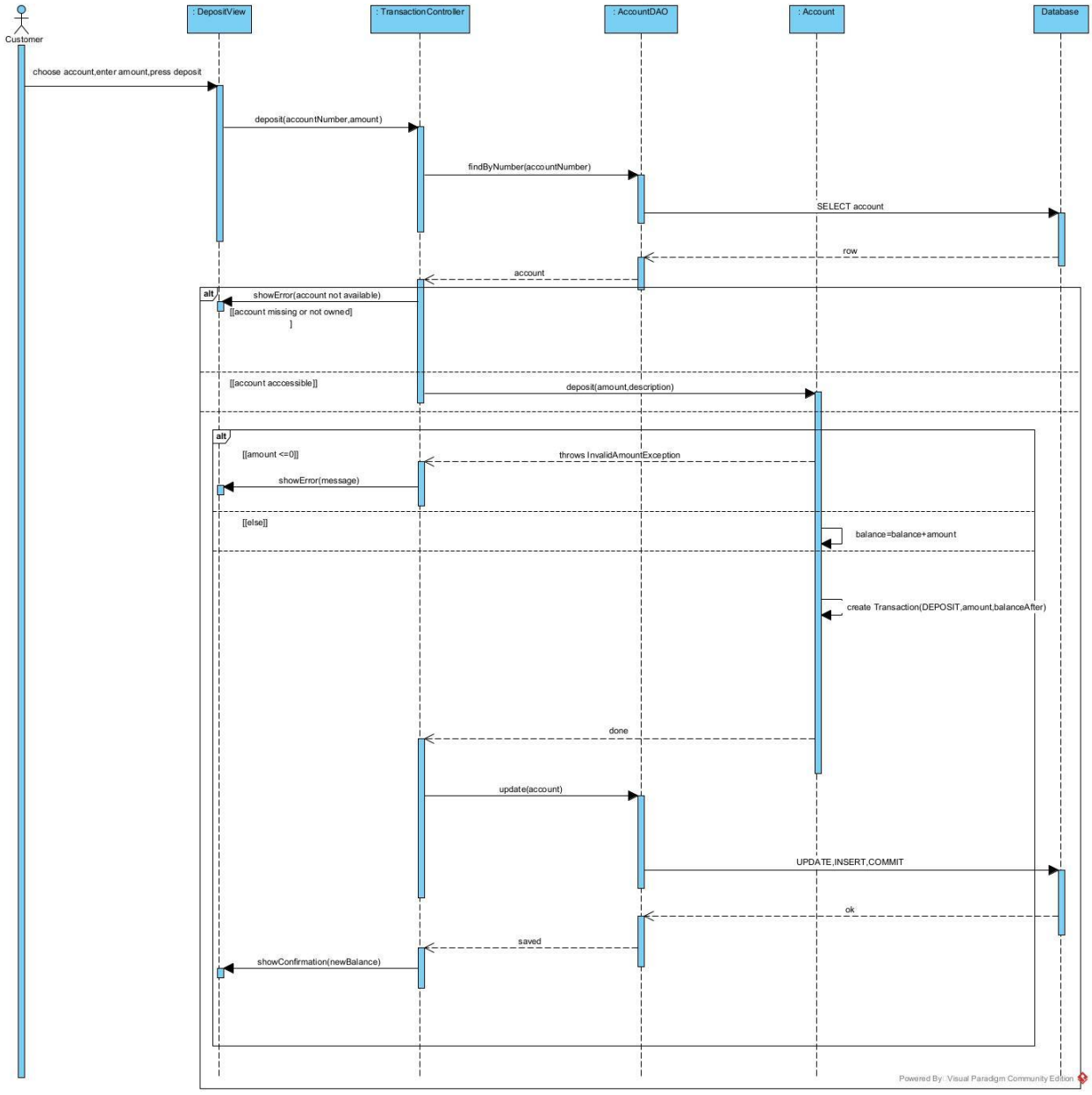
Activity diagram



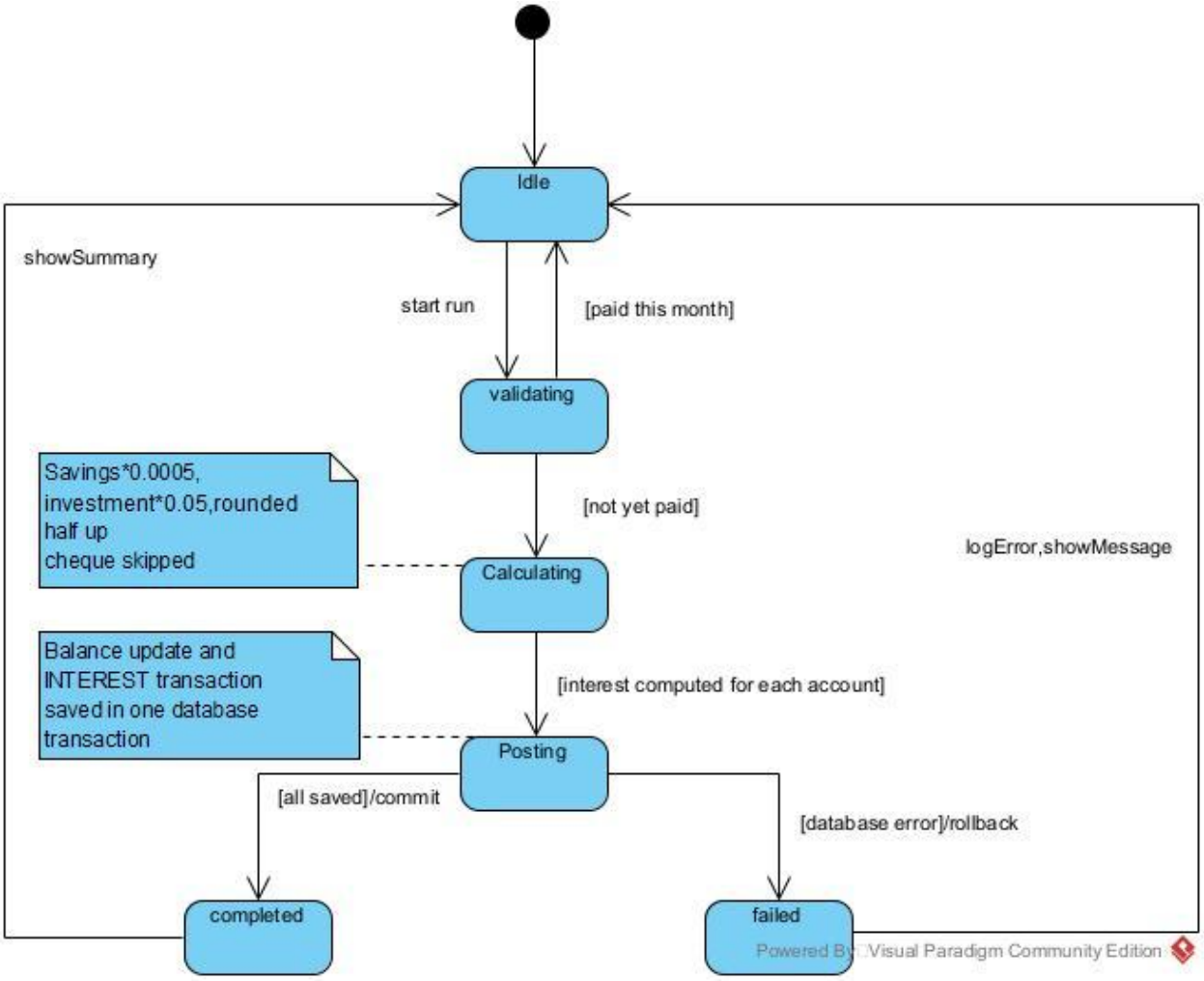
Sequence diagram-Login



Sequence diagram-Deposit funds



State Machine Diagram


Idle
showSummary
start run
[paid this month]
validating
Savings*0.0005,
investment*0.05,rounded
half up
cheque skipped
[not yet paid]
logError,showMessage
Calculating
Balance update and
INTEREST transaction
saved in one database
transaction
[interest computed for each account]
Posting
[all saved]/commit
[database error]/rollback
completed
failed
Powered By Visual Paradigm Community Edition